ARNAB KANTI JANA

akj.chs2014@gmail.com

(+91) 9734490372

Personal Webpage

Google Scholar

0009-0003-9115-5293

Indian

Present Research Focus

- **Core Physics:** Experimental investigation of strongly correlated electron systems, focusing on quantum magnetism, spin dynamics, and complex magnetic ground states in low-dimensional materials exhibiting geometric and bond-dependent magnetic frustration.
- **Structural Property:** Investigating the fundamental physics driven by the competition among electron correlation, spin-orbit coupling, and crystal field effects to elucidate the electronic states and origin of local magnetic moments in novel transition metal compounds.
- **Methodological Approach:** Correlating macroscopic thermodynamic and transport measurements with microscopic magnetic and structural states. This relies on a comprehensive workflow-spanning solid-state material synthesis, spectroscopic determination of electronic structure (XAFS, XPS), and advanced large-scale facility probes (neutron scattering, muon spin relaxation, and synchrotron X-rays).

Research Publications

1. **A. Jana**, A. Bhattacharyya, D. Das, and D. Adroja, "Exploring the disordered magnetic ground state and spin dynamics in the honeycomb oxide $Na_3Cu_2SbO_6$," *Journal of Alloys and Compounds*, vol. 1034, p. 181 080, 2025. DOI: 10.1016/j.jallcom.2025.181080
2. **A. Jana**, A. Bhattacharyya, C. Ritter, D. Adroja, and A. Hillier, "Successive antiferromagnetic and ferromagnetic phase transitions in $PrRu_2Al_2B$: A comprehensive study of magnetic transitions," *Journal of Alloys and Compounds*, vol. 1022, p. 179 756, 2025. DOI: 10.1016/j.jallcom.2025.179756
3. A. Bhattacharyya, D. Adroja, **A. Jana**, K. Panda, P. Ferreira, Y. Zhao, T. Ying, H. Hosono, T. Dorini, L. Eleno, P. Biswas, G. Stenning, R. Tripathi, and Q. Y, "Exploring superconductivity in $Ba_3Ir_4Ge_{16}$: Experimental and theoretical insights," *Journal of Alloys and Compounds*, vol. 978, p. 173 374, 2024. DOI: 10.1016/j.jallcom.2023.173374

Hands on Expertise

Synthesis	Solid-state reaction, Arc-furnace, Optical Floating Zone.
Structural Analysis	Lab based XRD (Bruker D8 Advance), Synchrotron XRD (RRCAT, BL12, Indus-2), SEM-EDS.
Magnetometry & Transport	DC and AC magnetization (MPMS3); heat capacity and electrical transport (PPMS - Quantum Design).
Spectroscopy	XPS, Raman scattering, UV-VIS-NIR, and Photoluminescence, and Dielectic.
Microscopic Local Probes	Muon Spin Relaxation (XX), Neutron Diffraction (XX), Inelastic Neutron Scattering (XX), X-ray Absorption Fine Structure (RRCAT, BL09, Indus-2).
Software	FullProf, JANA, WiMDA, Mantid, Demeter (Athena, Artemis, Hephaestus), Python, Fortran, Mathematica, VESTA, OriginLab, LaTeX.

Large-Scale Facility Experience & Accepted Proposals

ISIS Neutron and Muon Source (STFC)

2610315, MERLIN — Probing Magnetic excitation and Crystal Field States in the Quasi-2D Honeycomb layered oxide $Na_3Cu_2SbO_6$. (Awarded: 1 day)

2310347, EMU — Exploring the contribution of spin fluctuations in the formation of quantum spin liquid like state in $KLnSe_2$ (Ln = Ce and Pr). (Awarded: 4 days)

2220325, MUSR — Investigation of superconductivity in thiospinel compound $CuCo_2S_4$ using muon spin relaxation and rotation. (Awarded: 2 days)

Paul Scherrer Institute (PSI)

20211455, GPD — Search for Kitaev Spin Liquid in 3d Transition Metal Oxide. (Awarded: 2 days)

20221271, GPS — Investigation of the quantum disordered magnetic ground state in $RbCeO_2$ by muon. (Awarded: 1 day)

20221274, GPS — Study of spin fluctuating states in honeycomb layered oxide $Cu_3M_2SbO_6$ (M = Co and Ni) using muon spin relaxation. (Awarded: 1 day)

20221276, GPD — Investigation of superconducting dome on four-layer-type bismuth chalcogenides : $LaOPbBiS_3$ and $La_2O_2Bi_3AgS_6$ by muon. (Awarded: 5 days)

Deutsches Elektronen-Synchrotron (DESY)

20220829, PETRA III: P24 — Understanding the role of structural disorder at ambient pressure to the observed QSL like state in $AYbSe_2$ (A = Rb and Cs): A temperature dependent X-ray study. (Awarded: 3 days)

20220336, PETRA III: P24 — Understanding the role of structure in the observed quantum spin liquid-like state in $KLnSe_2$ (Ln = La, Ce, Pr, and Nd). (Awarded: 2 days)

Japan Proton Accelerator Research Complex (J-PARC)

2022B, Muon D1 — Investigation of the magnetic ground state in Bilayer Square Lattice $Tb_2SrAl_2O_7$ having magnetic frustration. (Awarded: 2 days)

Raja Ramanna Centre for Advanced Technology (RRCAT)

6025, INDUS-II BL09 — Investigating the oxidation state of Cu and its local environment in $Na_3Cu_2SbO_6$. (Awarded: 2 days)

5525, INDUS-I BL02 — Study of averaged electronic structure of $Na_3Cu_2SbO_6$. (Awarded: 7 days)

5524, INDUS-II BL12 — Investigation of structural change/distortion prior to the AFM order in $Na_3Co_2SbO_6$. (Awarded: 2 days)

5383, INDUS-II BL12 — Investigation of structural phase transition in $Na_3Cu_2SbO_6$. (Awarded: 2 days)

Conference Presentations & Invited Talks

- **Invited Talk:**

  2025 *Novel Emergent Properties of Quantum Materials*, FEL Colloquium, Raja Ramanna Centre for Advanced Technology (RRCAT), Indore, India.

- **Contributed Talks:**

  2022 *Magnetic Excitations In a Novel 1D, Spin-1/2 Chain Magnet: $CuSb_2O_4$*, 66th DAE Solid State Physics Symposium, BIT-Mesra, India.

- **Poster Presentations:**

2025 *Exploring the disordered magnetic ground state and spin dynamics in the honeycomb oxide* $Na_3Cu_2SbO_6$, Synchrotron X-ray User Outreach Workshop (DST, India-DESY, Germany Collaboration), IIT-BHU, Varanasi, India.

2022 *Investigation of the superconducting gap structure in the layered caged compound* $Ba_3Ir_4Ge_{16}$ *using muon-spin relaxation and rotation*, International Conference on Spectroscopies of Novel Superconductors (SNS), IISC Bangalore, India.

2022 *Large Magnetocaloric Effect in* $Ho_{15}Si_9C$, Conference on Emergent phenomena in Quantum MATerials (E-QMAT), IIT-Roorkee, India.

- **Schools & Workshops Attended:**

2019 Interactive workshop on "Experimental physics" (IWEP-2019), Sponsored by West Bengal state council of Science & Technology.

2018 C.K. MAJUMDAR memorial summer workshop, S.N. Bose national centre for basic science.

Awards & Achievements

2021 – 2026 **DST-INSPIRE Fellowship**, Department of Science and Technology (DST), Govt. of India.
5-year national fellowship for doctoral research. Awarded as Senior Research Fellow (SRF) from 2024–2026 and Junior Research Fellow (JRF) from 2021–2023.

2023 **DST-DESY International Travel Grant**, JNCASR, India.
Full financial support awarded to conduct synchrotron experiments at PETRA III, DESY, Germany (Proposal IDs: I-20220336 & I-20220829).

2018 **Gold Plated Medal for 1st Position in B.Sc. Physics**, Prabhat Kumar College, India.
Awarded for securing the highest academic standing in the graduating class.

2015 **DST-INSPIRE Scholarship for Higher Education (SHE)**, Department of Science and Technology (DST), Govt. of India.
National scholarship awarded for pursuing higher education in basic sciences.

2012 **First Prize, District Level Science Fair**, Digha Science Centre & National Science Camp, India.